ModelIt!

Student Activity Pack

Metabolic Engineering for Biofuels

Pathway Optimization and Flux Balancing in Microbial Fuel Production

Grade 09L3 | NGSS HS-LS1-5, HS-ETS1-3 | Energy and Matter / Stability and Change

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **What do you think happens inside a yeast cell when it ferments sugar into alcohol — and how is that related to making fuel?**

2. **Why do you think we can't just grow corn and turn it all into ethanol to replace gasoline?**

3. **Draw a diagram showing how you think a cell converts sugar into a useful chemical product — where does the carbon go?**

4. **What does 'metabolic engineering' mean to you and how might it differ from genetic engineering?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Substrate Concentration | Enzyme Activity | Pathway Flux | Cofactor Availability | Metabolic Burden | Cell Growth Rate | Product Yield | Byproduct Accumulation | Fermentation Efficiency

EXTERNAL	INTERNAL

? Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Baseline Fermentation

Standard substrate, moderate enzyme expression

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Maximum Expression

All pathway enzymes at maximum expression

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Balanced Optimization

Tuned expression with cofactor engineering

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Product Toxicity Threshold — The concentration of biofuel product at which it begins to damage the producing cell's membrane and proteins — ethanol tolerance in yeast is 12-15% while butanol kills most bacteria above 2%, making product toxicity one of the most critical limitations in biofuel fermentation

Feedstock Flexibility — The range of carbon substrates the engineered organism can utilize — strains that can co-ferment glucose AND xylose from lignocellulosic biomass access 40% more available carbon than glucose-only strains, dramatically improving process economics

Genetic Stability Index — The probability that the engineered production pathway maintains full function over extended fermentation runs — cells under high Metabolic Burden face selective pressure to mutate and inactivate costly pathway enzymes, causing production to decline over time

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

We burn 100 million barrels of oil every day. Can we engineer microbes to brew fuel instead of drilling for it — and can we make it economically competitive? Use evidence from your simulation to support your answers.

1. **How does your model demonstrate the fundamental trade-off between Cell Growth Rate and Product Yield — and why can't you simply maximize both?**

2. **Why is Cofactor Availability often called the 'silent killer' of engineered metabolic pathways?**

3. **What did you learn about the relationship between Enzyme Activity levels and Metabolic Burden — why doesn't 'more enzyme' always mean 'more product'?**

4. **How would adding Product Toxicity Threshold change your optimization strategy for Fermentation Efficiency?**

5. **What are the environmental justice implications of siting large biofuel fermentation facilities — who benefits and who bears the costs?**

8 STEM Challenge: Design a Next-Generation Biofuel Production Strain

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YOUR ENGINEERING MISSION

Design an engineered microbial strain and fermentation process that produces advanced biofuel at commercially competitive yields, titers, and productivities.

THE SCENARIO

A renewable energy startup has secured $50 million in venture capital to develop a microbial biofuel that can replace jet fuel. Current petroleum-based jet fuel costs $0.60 per liter to produce. Your metabolic engineering team must design a microbial strain and fermentation process that produces a drop-in jet fuel replacement at competitive cost — which means achieving high Product Yield, high titer (above 50 g/L), and high volumetric productivity (above 1 g/L/hr).

Engineering Design Thinking

1. WOULD YOU USE A BATCH, FED-BATCH, OR CONTINUOUS FERMENTATION PROCESS — AND HOW DOES THIS CHOICE AFFECT SUBSTRATE CONCENTRATION MANAGEMENT?

2. HOW WOULD YOU ENGINEER THE STRAIN TO TOLERATE HIGH PRODUCT CONCENTRATIONS THAT ARE TYPICALLY TOXIC TO THE CELL?

3. WHAT GENETIC STABILITY STRATEGIES WOULD YOU IMPLEMENT TO PREVENT THE PRODUCTION PATHWAY FROM MUTATING OUT DURING LONG FERMENTATIONS?

4. HOW DOES THE CARBON FOOTPRINT OF YOUR BIOFUEL COMPARE TO PETROLEUM WHEN YOU ACCOUNT FOR SUBSTRATE PRODUCTION, FERMENTATION ENERGY, AND DOWNSTREAM PROCESSING?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Metabolic Engineers design and optimize microbial production pathways for fuels, chemicals, and pharmaceuticals. They work for biotech companies (Amyris, Ginkgo Bioworks, LanzaTech, Genomatica), renewable energy startups, and national laboratories, earning $90,000-$175,000/year. Fermentation Scientists who scale up lab strains to industrial production earn $80,000-$150,000/year.